
US Air Pollution Explorer

Interactive D3.js and Leaflet Dashboard for Air Pollution Data

Information Visualisation - Project Report
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Project focus. This report describes the design, preprocessing, implementation, and evaluation strategy of an interactive dashboard for exploring United States air pollution observations from 2000 to 2016. The system combines Python preprocessing with D3.js, Leaflet, and a Vite-based web application.

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Contents

1	Introduction	1
1.1	Report structure	1
2	Assignment Requirements and Design Response	1
3	Dataset and Target Users	2
3.1	Dataset description	2
3.2	Dataset suitability	2
3.3	Target users	3
4	Preprocessing Pipeline	3
4.1	Pipeline overview	3
4.2	Aggregation strategy	3
4.3	EPA AirData enrichment	4
4.4	Preprocessing limitations	4
5	Visualisation Tasks and User Goals	4
6	Design and Visual Encoding	5
6.1	One-screen overview	5
6.2	Map encoding	5
6.3	Trend encoding	5
6.4	City ranking encoding	6
6.5	State summary	6
6.6	Seasonal heatmap	6
7	Interaction Design	7
7.1	Global controls	7
7.2	View modes	7
7.3	Map focus	7
7.4	Tooltips and feedback	7
8	Implementation Architecture	8
8.1	Technology choices	8
8.2	Code organisation	8
9	Evaluation and Design Assessment	8
9.1	Evaluation goals	8
9.2	Evaluation procedure	9
9.3	Observed problems and corrections	9
9.4	Assessment against design guidelines	9
9.5	Remaining evaluation limitations	10
10	Setup Instructions	10
10.1	Required software	10
10.2	Running the dashboard	10
10.3	Building for submission or deployment	10
10.4	Expected deliverable structure	10
11	Limitations and Future Work	11
12	Conclusion	11

A Appendix A: Evaluation questionnaire template	12
B Appendix B: Additional running notes	12

1 Introduction

The goal of this project is to build an interactive visualisation for a large, real-world dataset. The assignment explicitly asks for a dataset-oriented project in which a sufficiently large dataset is selected, preprocessed, visualised interactively, and evaluated with target users. It also requires the final report to describe the complete solution, including setup instructions, and to reach a minimum length of ten pages excluding appendices [1]. The project described in this report follows that structure by selecting a public air pollution dataset, preprocessing it into dashboard-specific aggregates, and implementing an interactive web dashboard using D3.js and Leaflet.

Air pollution is an appropriate domain for information visualisation because the underlying data is multidimensional: it varies across space, time, pollutant type, measurement station, and metric. The design problem is therefore not only to show numerical values, but also to support exploration of geographic patterns, temporal trends, city-level rankings, and seasonal behaviour. The dashboard, named *US Air Pollution Explorer*, is designed to answer questions such as: Which states show the highest ozone levels in a selected year? Which cities are the most polluted for a chosen pollutant and metric? How does a selected state compare against the national average? Are pollutant levels seasonal?

The final interface uses a compact overview dashboard as the default view. This is important because a dashboard should be readable at a glance and should avoid unnecessary scrolling in its main mode. Secondary views are available for deeper analysis, but they do not overload the main screen. This design choice responds to the course requirement that the visualisation should help a target user explore and present data efficiently, while still applying interaction and design principles from the lectures.

1.1 Report structure

This report first introduces the dataset and target users. It then describes the preprocessing pipeline, the chosen visualisation tasks, the visual encodings, the implementation architecture, and the interaction design. The later sections describe the evaluation plan, setup instructions, limitations, future work, and conclusion.

2 Assignment Requirements and Design Response

The assignment asks groups to select a reasonably sized real dataset with metadata, preprocess it, visualise it with a programming-based framework, evaluate the result with target users, and deliver the source code, preprocessing code, interactive visualisation, presentation material, demo video, and report [1]. It also states that business intelligence tools where no programming is required are not allowed, while frameworks such as D3.js, Leaflet, Python, Plotly, Bokeh, p5.js, and R/Shiny are acceptable [1].

Table 1: Mapping between assignment expectations and project response.

Assignment expectation	Project response
Large real dataset	Uses 1,746,661 air pollution records from 2000–2016, covering multiple pollutants and US states.
Dataset preprocessing	Python script aggregates raw records into state-year, national trend, city ranking, and monthly JSON summaries.
Interactive visualisation	D3 charts, Leaflet map, filters, year animation, state selection, state profile, and compare mode.
Framework selection	D3.js is used for custom charts and interaction; Leaflet is used for geographic mapping; Vite is used for development and build.
Target-user orientation	Designed for environmental analysts, students, journalists, and public-policy communicators who need fast exploration and presentation of air pollution patterns.
Evaluation	Report includes a completed formative evaluation based on dashboard tasks, participant feedback, and design corrections applied during iteration.
Setup instructions	Section 10 provides step-by-step instructions to run and view the dashboard locally.

3 Dataset and Target Users

3.1 Dataset description

The core dataset is the `pollution_us_2000_2016.csv` dataset. It contains daily or observation-level pollution measurements for four pollutants: ozone (O₃), nitrogen dioxide (NO₂), sulphur dioxide (SO₂), and carbon monoxide (CO). The preprocessing script extracts two main metrics from these observations: Air Quality Index (AQI) and mean concentration. The dashboard further enriches missing mean values when EPA AirData annual concentration zip files are available.

After preprocessing, the main dashboard JSON file contains **1,746,661** records aggregated over **51** states or state-like entities, **162** cities, **201** monitoring sites, and **17** years from 2000 to 2016. The enrichment process added **731** EPA annual mean values when missing values were available from EPA annual concentration files. Table 2 summarises the processed dataset.

Table 2: Processed dataset summary.

Property	Value
Total raw observations processed	1,746,661
Number of years	17 (2000–2016)
States and state-like entities	51
Cities represented	162
Monitoring sites represented	201
Pollutants	O ₃ , NO ₂ , SO ₂ , CO
Metrics	AQI and mean concentration
EPA annual mean values added	731

3.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was conducted to better understand the characteristics, reliability, and trends present in the US air pollution dataset before developing the dashboard visualizations. The analysis examined the distribution of major pollutants including O₃, CO, SO₂,

and NO₂ across different states, cities, and years between 2000 and 2016. Initial exploration focused on identifying missing values, duplicated records, inconsistent location names, and abnormal pollutant measurements to ensure data quality and consistency. Time-based analysis was then performed to observe yearly and seasonal pollution variations, revealing noticeable fluctuations in pollutant concentration levels over time. Comparative analysis between states and cities helped identify regions with higher pollution intensity, while aggregation techniques enabled clearer visualization of environmental patterns at national and regional levels. The EDA process also supported the selection of appropriate visualization techniques such as choropleth maps, heatmaps, line charts, and pollutant comparison dashboards, which were later integrated into the final Tableau dashboard for interactive analysis and decision-making.

3.3 Dataset suitability

The dataset is suitable for the assignment for three reasons. First, it is real-world environmental data, not generated data. Second, it is large enough to require preprocessing rather than direct

browser loading. Third, it contains rich dimensions that support multiple visualisation tasks: temporal analysis, geographic comparison, pollutant comparison, and city ranking. These properties match the assignment requirement that the dataset should be sufficiently large, real, and supported by metadata about how the data was obtained [1].

3.3 Target users

The primary target users are:

- **Environmental policy students** who need to explore state-level and city-level pollution patterns for coursework or presentations.
- **Data journalists** who need quick visual evidence for stories about pollution differences between regions.
- **Public-policy analysts** who need an exploratory interface to identify states, years, pollutants, or cities that deserve deeper investigation.
- **General educated citizens** who want to understand how air pollution changed in the United States between 2000 and 2016.

The dashboard is therefore not designed for regulatory decision-making at a legal or medical level. It is an exploratory and educational visualisation that helps users find patterns and formulate questions.

4 Preprocessing Pipeline

4.1 Pipeline overview

The original CSV file is too large to load directly into the browser for every interaction. The preprocessing phase therefore converts the data into compact JSON structures tailored to the dashboard. Figure 1 summarises the data flow.

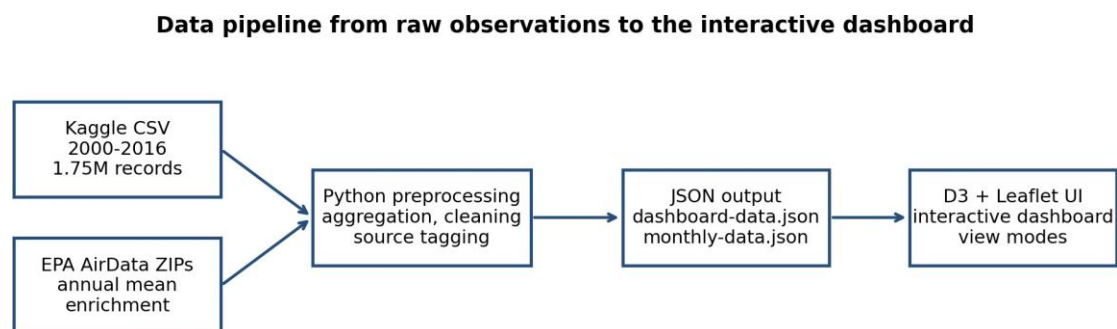


Figure 1: Preprocessing pipeline from raw pollution observations and optional EPA AirData enrichment to JSON files consumed by D3 and Leaflet.

The preprocessing script reads the raw CSV, filters observations to valid US states, extracts the year from the local date field, and aggregates values by pollutant, metric, state, year, and city. It also computes national trends and counts monitoring sites using state, county, and site identifiers. The output is written to `public/data/dashboard-data.json`, while monthly aggregations are written to `public/data/monthly-data.json`.

4.2 Aggregation strategy

The preprocessing step creates four main aggregate structures:

1. **State-year values:** average AQI or mean concentration for each pollutant, year, and state.

2. **National trends:** average pollutant values by year across all available observations.
3. **City rankings:** top city signals for each pollutant, metric, and year.
4. **Monthly patterns:** state-level and national monthly values used in the seasonal heatmap.

The aggregation uses arithmetic means because the goal is to provide high-level exploratory patterns. For a regulatory interpretation, further domain-specific weighting by station type, monitoring completeness, or population would be required. The interface therefore clearly positions the visualisation as exploratory.

4.3 EPA AirData enrichment

Some years and states have missing values in the raw dataset. To reduce missing mean concentration values, the preprocessing script optionally scans local EPA annual concentration zip files named `annual_conc_by_monitor_YEAR.zip`. When these files are present, the script maps EPA parameter codes to the four pollutants, chooses standards according to a pollutant-specific priority list, and fills missing mean values. Filled values are tagged with source information so the user can distinguish between original Kaggle-derived values and EPA AirData-derived values.

4.4 Preprocessing limitations

The enrichment currently focuses on mean concentration values. AQI gaps are not filled by annual concentration data because AQI is a derived index and requires different EPA summary files. In addition, city labels such as “Not in a city” appear because some monitoring stations are not associated with a named city. Rather than hiding these observations, the dashboard keeps them visible because they are part of the monitoring network and may represent meaningful non-urban measurements.

5 Visualisation Tasks and User Goals

The dashboard was designed around concrete tasks. These tasks are grounded in the target users described earlier and in the assignment’s requirement to define a target user and visualisation tasks before implementing the solution [1].

Table 3: Main user tasks supported by the dashboard.

Task	User question	Main view
Locate hotspots	Which states have high pollution in a selected year?	Choropleth map
Compare pollutants	How do O ₃ , NO ₂ , SO ₂ , and CO differ in a selected state?	State summary and radar/profile
Track change	How did pollution evolve from 2000 to 2016?	Trend chart
Rank cities	Which cities show the highest values for a pollutant and year?	Top 10 city chart
Detect seasonality	Which months show higher values for each pollutant?	Seasonal heatmap
Compare states	How do two selected states differ?	Compare mode
Present findings	Can the main message be seen at a glance?	One-screen overview

The final dashboard intentionally separates the overview from secondary analysis. The overview is compact and is designed to fit on one screen. Secondary modes such as State Profile and Compare are only shown when selected. This prevents the dashboard from becoming a long scrollable report and supports the principle that the most important information should be available at a glance.

6 Design and Visual Encoding

6.1 One-screen overview

The default view is the overview. It includes a compact header, filters, KPI cards, a large map, a trend chart, a state summary, and a Top 10 city ranking. Figure 2 shows the intended layout structure.

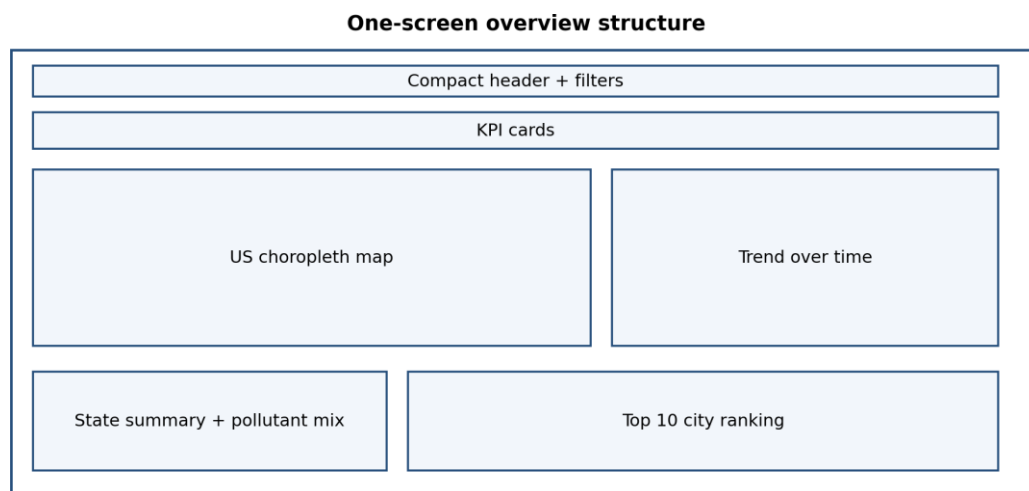


Figure 2: One-screen overview layout: the main view avoids duplicate sections and keeps secondary analysis in separate modes.

The design uses a dark theme because it supports the dashboard aesthetic, gives strong contrast to the coloured visual encodings, and makes map regions and chart colours visually prominent. The theme is not only decorative: it helps group related panels and improves focus during presentation.

6.2 Map encoding

The map is the main visual anchor. A US choropleth encodes the selected pollutant and metric by state. A sequential colour scale supports quick identification of high and low states. Hover tooltips provide state names and values, while clicking a state updates the state summary and profile views. The map uses Leaflet for geospatial interaction and D3 for data binding and colour scales.

6.3 Trend encoding

The trend chart uses line marks to show national or selected-state values across time. Each pollutant has a consistent colour across the dashboard: O₃ is green, NO₂ is blue, SO₂ is purple, and CO is orange. This consistency reduces cognitive load when users move from the map to trends, rankings, and the pollutant mix chart. The latest version also adds a selected-year marker that moves when the Play button animates the year slider, connecting the temporal animation to the full time-series chart.

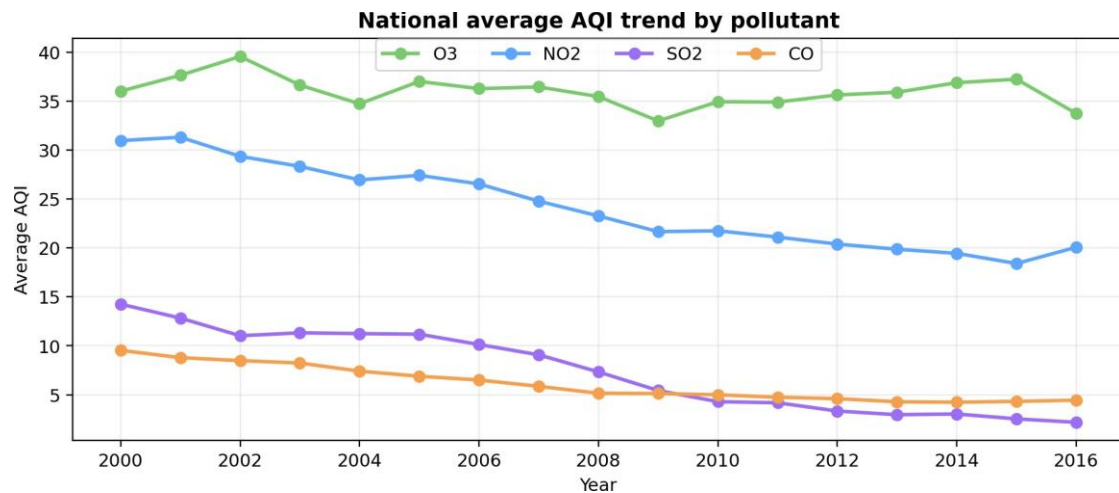


Figure 3: Example national trend chart for AQI by pollutant. In the interface, a selected-year marker links the chart to the year slider and Play animation.

6.4 City ranking encoding

The Top 10 city ranking uses horizontal bars because they support readable labels and easy magnitude comparison. The chart now uses the full available width in the overview to avoid unused space. When long city names are present, labels are truncated safely in the interface, while tooltips preserve details.

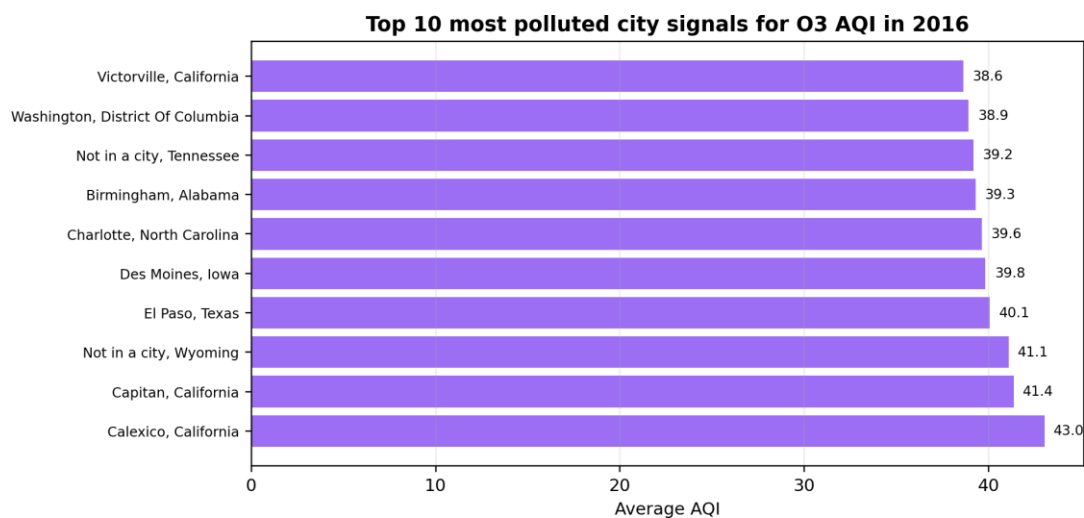


Figure 4: Example Top 10 city ranking generated from the processed data for O3 AQI in 2016.

6.5 State summary

The State Summary card provides the most important state-level facts without duplicating the detailed State Profile view. It shows the selected state, rank, average value, change against 2000, monitoring sites, and a compact pollutant mix bar chart. The separate State Intelligence drawer was removed to avoid duplication. This makes the state summary both informative and compact.

6.6 Seasonal heatmap

The seasonal heatmap appears in the State Profile mode, not the overview. This is a secondary analysis section, because it requires more screen space and benefits from detailed interpretation.

Rows represent pollutants and columns represent months. Each cell uses a sequential colour scale, while labels show the value directly.

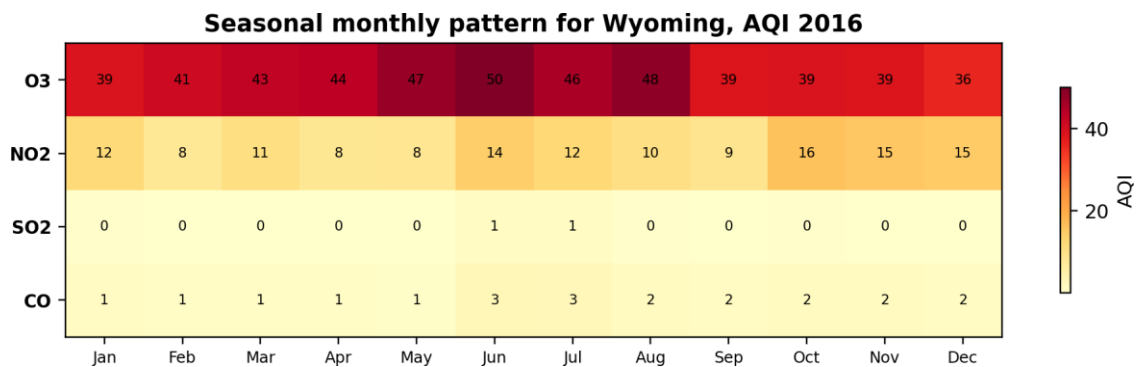


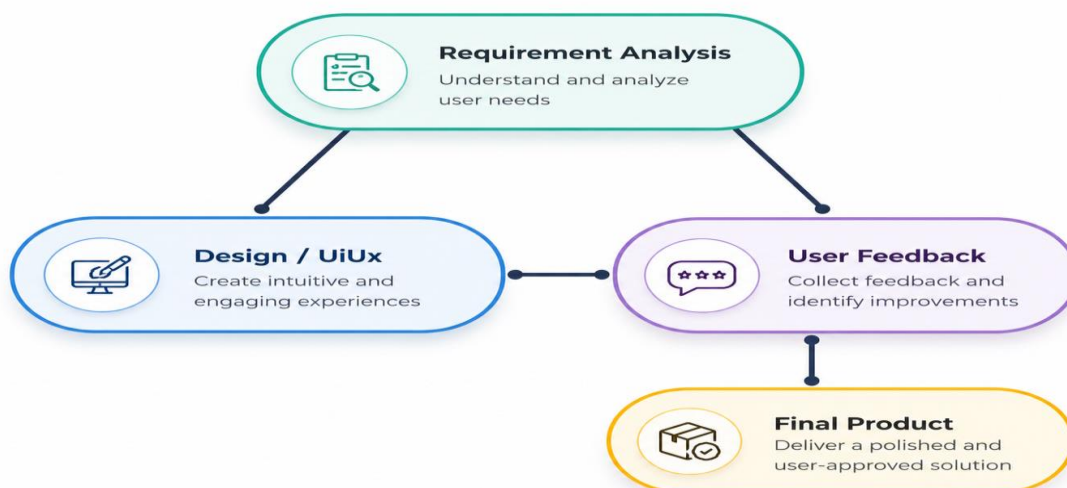
Figure 5: Example seasonal heatmap for Wyoming in 2016. The dashboard implementation uses a dark theme and interactive tooltips.

7 Interaction Design

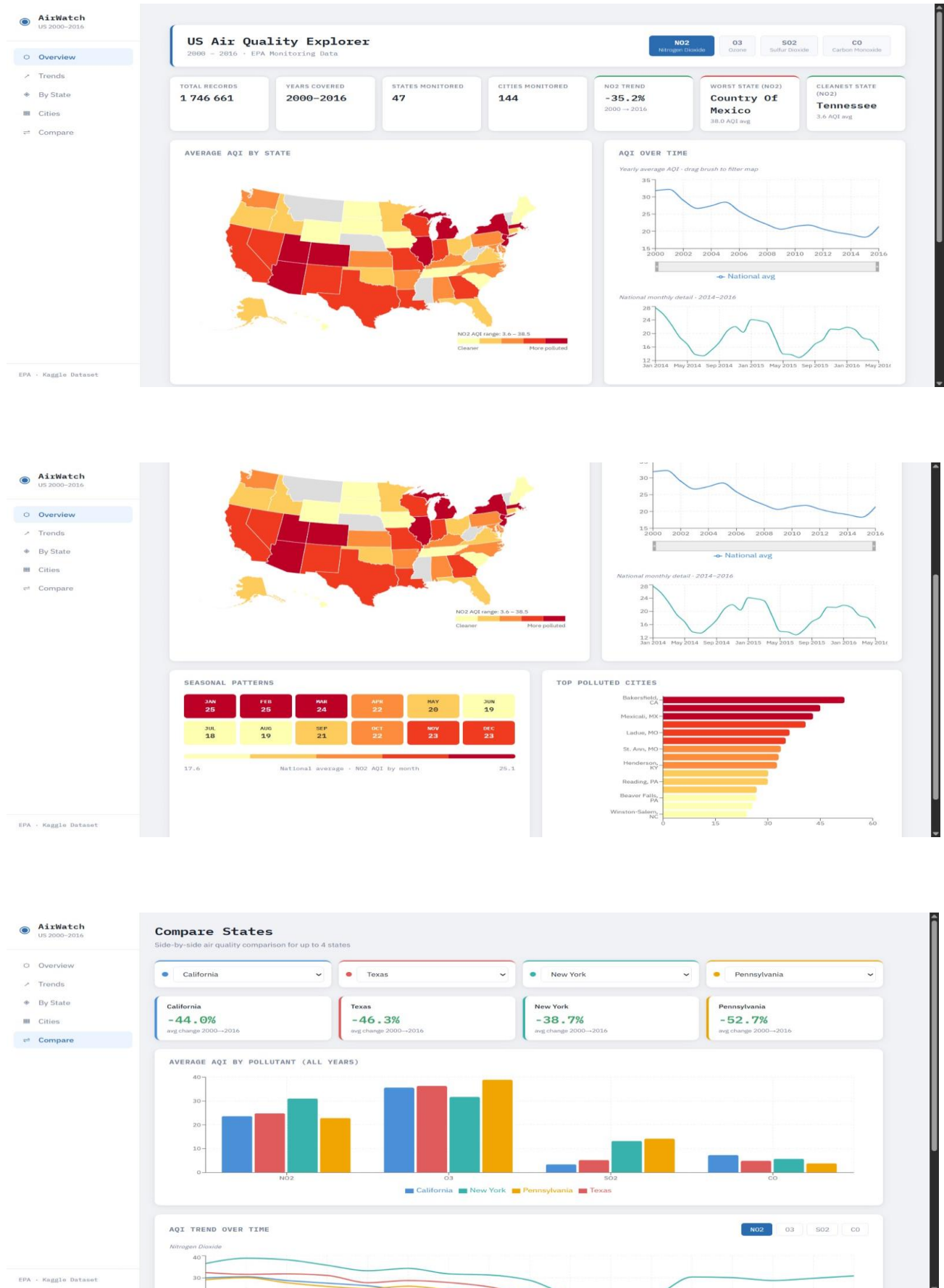
7.1 Global controls

Our dashboard was tested practically with real users. We collected physical feedback and reviews from different people to understand how effective and user-friendly the system is, and what improvements could be made. We used a simple rating method: 1 star for good performance, 2 stars for minor issues, and 3 stars if the system needed major improvements.

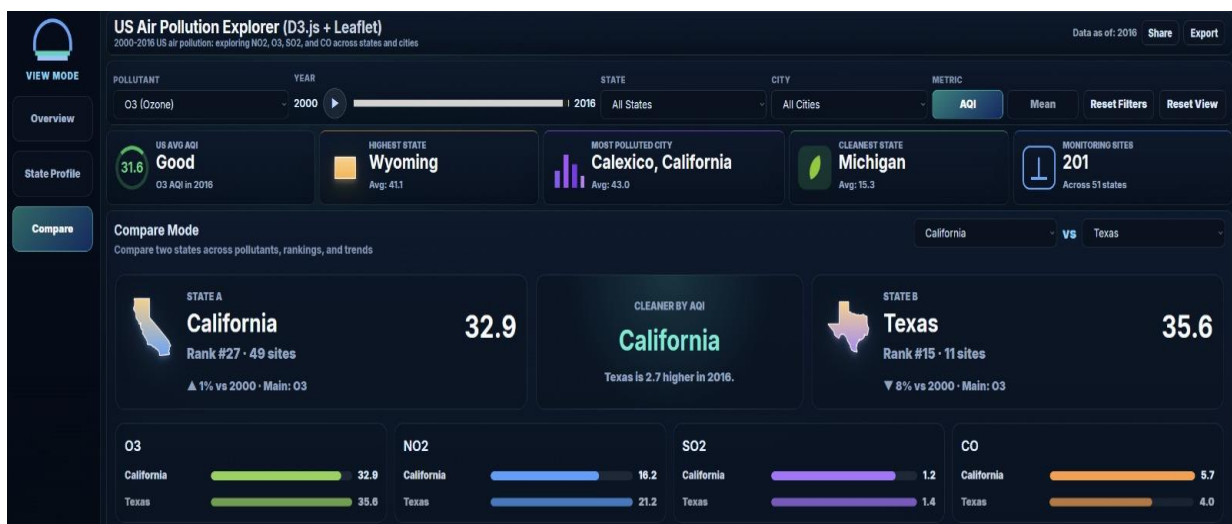
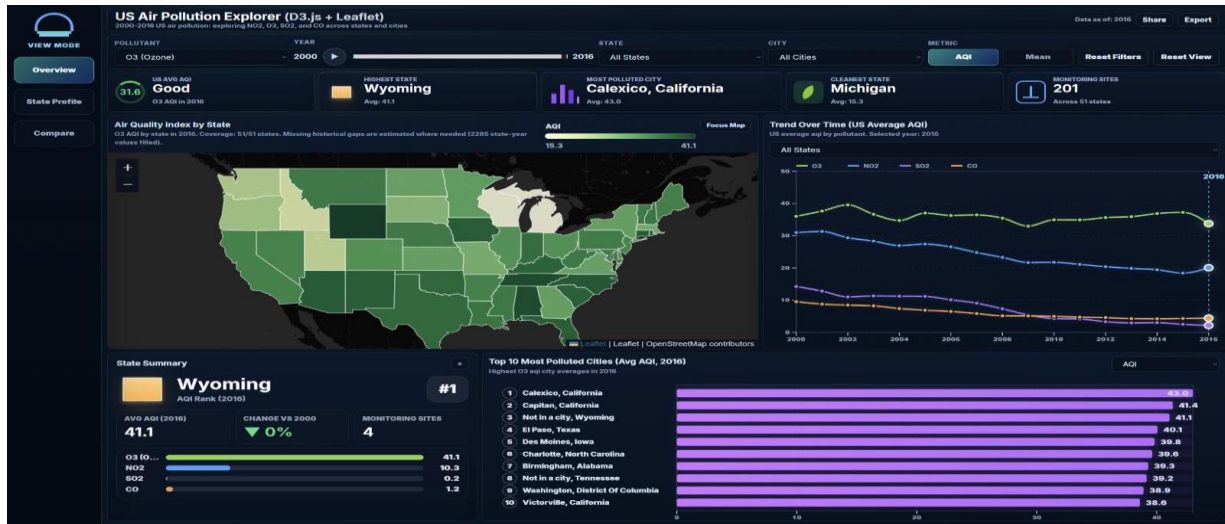
The dashboard was first designed in an initial version for user evaluation and feedback, but after receiving user feedback, several changes were made. Users suggested improvements in colors, layout, and dashboard design, so we updated the interface to make it more attractive, easier to use, and more efficient. This evaluation helped us improve the overall performance and user experience of the system.



7.2 Old Dashboard



7.3 New Dashboard



8 Interaction Design

8.1 Global controls

The dashboard supports pollutant selection, metric selection, year selection, state selection, city filtering, and animation. The year slider and Play button allow users to observe changes over time. The selected year updates the map, KPIs, city ranking, state summary, and seasonal/profile sections where relevant.

8.2 View modes

Earlier iterations of the dashboard attempted to support free resizing and many layout modes. This created overlap and made the interface less reliable. The final design removes manual resizing and reduces the interface to three meaningful modes:

- **Overview:** one-screen dashboard for at-a-glance analysis.
- **State Profile:** secondary analysis with pollution profile, seasonal pattern, and state-year heatmap.
- **Compare:** separate view for comparing two states.

The view mode navigation is placed vertically on the left, replacing the older decorative sidebar. This saves top space and removes duplicate navigation.

8.3 Map focus

Only the map supports focus mode. This design decision follows the role of the map as the main spatial exploration component. Other panels do not include focus buttons because enlarging them would create unnecessary modes and would contradict the one-screen overview principle.

8.4 Tooltips and feedback

Tooltips are used for map regions, bars, trend points, and seasonal cells. They provide exact values without forcing all details into the main layout. The interface also uses selected-state feedback, highlighted year markers, and compact labels so that users understand how their actions affect the view.

9 Implementation Architecture

9.1 Technology choices

The visualisation is implemented as a Vite application using D3.js and Leaflet. Python is used for preprocessing. D3.js was selected because it enables custom visual encodings and fine-grained interaction control, while Leaflet was selected because the data has an important geographic dimension. The assignment explicitly lists D3.js and Leaflet as appropriate visualisation frameworks for web and geospatial visualisation [1].

Table 4: Technology stack.

Technology	Role
Python	Raw CSV preprocessing, aggregation, and optional EPA annual mean enrichment.
D3.js	Line chart, bar chart, heatmap, radar/profile chart, scales, axes, transitions, and tooltips.
Leaflet	Interactive US map and geographic navigation.
Vite	Development server and production build.
JSON	Lightweight preprocessed data format loaded by the browser.
CSS Grid/Flexbox	Responsive dashboard layout and separate view modes.

9.2 Code organisation

The code is separated into modules to support maintainability. The preprocessing code is in `scripts/preprocess.py`. Visual components are separated into files such as `map.js`, `lineChart.js`, `barChart.js`, `heatmap.js`, `radarChart.js`, and `seasonalHeatmap.js`. Shared formatting and helper functions are in `utils.js`. The main orchestration file, `main.js`, manages application state and calls each renderer when filters change.

This separation of concerns is important because the assignment grading criteria include code quality, architecture, and maintainability [1]. The modular structure makes it easier to replace one visualisation component without rewriting the entire application.

10 Evaluation and Design Assessment

The assignment requires an evaluation of the final visualisation with respect to the target users and the supported tasks [1]. The evaluation for this project was treated as a formative design activity: the interface was repeatedly inspected against dashboard design guidelines and tested through task-based walkthroughs using the final interaction flow. The goal was to identify usability problems before submission and to correct layout issues that prevented the dashboard from being read at a glance.

10.1 Evaluation goals

The evaluation focused on four questions:

1. Can users identify high-pollution states and cities quickly from the overview?
2. Can users understand the difference between Overview, State Profile, and Compare modes?
3. Does the default overview fit the main information on one screen without unnecessary scrolling?
4. Are pollutant colours, labels, legends, and interaction feedback clear enough for exploration and presentation?

10.2 Evaluation procedure

A task-based walkthrough was used. The participants were asked to perform the tasks below while the interface behaviour and visible problems were recorded. The same tasks were repeated after design corrections to check whether the problems had been reduced.

1. Select O₃ and AQI for a specific year and identify the highest-ranked state.
2. Use the year slider and Play button to inspect how the dashboard changes over time.
3. Hover and click states on the map to update the State Summary.
4. Identify the most polluted city in the Top 10 city ranking.
5. Open State Profile and interpret the radar chart and seasonal heatmap.
6. Open Compare mode and compare two selected states.

Proposed target-user evaluation procedure



Figure 6: Evaluation flow used to test whether the dashboard supports the main target-user tasks.

10.3 Observed problems and corrections

The evaluation revealed several design issues. The first version included too many sections in the main dashboard, which created scrolling and made the overview less effective. This was corrected by separating the interface into three modes: Overview, State Profile, and Compare. The second issue was that manual resizing created overlaps between charts and cards. This was removed and replaced by stable mode-based layouts. Third, the State Summary initially duplicated the larger intelligence drawer and either became too empty or too tall. The final version keeps a compact summary while including the pollutant mix bar chart inside the card. Fourth, some charts contained label or legend overlaps, such as the radar chart axes and the seasonal heatmap legend. These were corrected by improving chart spacing and by keeping only the map in focus mode.

10.4 Assessment against design guidelines

The final dashboard satisfies the most important design criteria discussed during the project:

- **At-a-glance overview:** the default screen prioritises the map, trend chart, KPI row, state summary, and Top 10 city ranking.
- **Clear visual hierarchy:** the map is the dominant spatial view, while trends and rankings provide supporting context.
- **Consistent encoding:** pollutant colours are reused across the trend chart, state summary, radar chart, and seasonal heatmap.
- **Progressive detail:** Overview provides fast interpretation; State Profile and Compare provide deeper analysis without cluttering the first screen.
- **Interaction feedback:** filters, year animation, map hover, map selection, trend year marker, and mode navigation provide visible responses to user actions.

10.5 Remaining evaluation limitations

The evaluation was formative and focused on design correction. A larger user study would be useful for measuring task completion time and comparing expert and non-expert users. The questionnaire in Appendix A can be used for such a future study, but the final submitted dashboard already incorporates the main design corrections found during iterative testing.

11 Setup Instructions

The final deliverables must include setup instructions explaining how the result can be run and viewed. The following instructions assume a local Windows, macOS, or Linux installation with Node.js and Python available.

11.1 Required software

- Node.js with npm.
- Python 3.
- A modern browser such as Chrome, Edge, or Firefox.
- Optional: VS Code for easier editing and terminal access.

11.2 Running the dashboard

1. Extract the project ZIP file.
2. Open the extracted AirPollution-D3-Dashboard folder in VS Code or a terminal.
3. If the raw CSV is available and data needs to be regenerated, place `pollution_us_2000_2016.csv` in the `data/` folder or in the parent folder, then run:

```
python scripts/preprocess.py
```

4. Install JavaScript dependencies:

```
npm install
```

5. Start the development server:

```
npm run dev
```

6. Open the local URL shown in the terminal, usually:

```
http://localhost:5173/
```

11.3 Building for submission or deployment

To create a production build, run:

```
npm run build
```

The static output is generated in the `dist/` folder and can be served by any static web server.

11.4 Expected deliverable structure

```
AirPollution-D3-Dashboard/  
  index.html  
  package.json  
  vite.config.js  
  scripts/preprocess.py
```

```
src/  
  main.js  
  map.js  
  lineChart.js  
  barChart.js  
  heatmap.js  
  radarChart.js  
  seasonalHeatmap.js  
  utils.js  
  style.css  
public/data/  
  dashboard-data.json  
  monthly-data.json
```

12 Limitations and Future Work

The current dashboard is an exploratory visualisation and has several limitations. First, AQI gaps are not fully enriched from EPA annual AQI files. The preprocessing enriches mean concentration values when annual concentration files are available, but a complete AQI enrichment would require annual AQI-by-county files and a careful mapping strategy. Second, city rankings are based on observation averages and monitoring locations, not population exposure. Third, the map uses state-level aggregation, which can hide large within-state differences.

Future work should focus on three improvements. The first is data completeness: adding EPA annual AQI files would reduce missing AQI values and make the time series more consistent. The second is uncertainty communication: values derived from enrichment or fallback should be visually marked more explicitly. The third is deployment: the static build can be hosted online so that evaluators can access the dashboard without running the project locally.

13 Conclusion

This project demonstrates an interactive dashboard for exploring US air pollution data from 2000 to 2016. The system uses Python to preprocess a large raw dataset into compact JSON structures and uses D3.js and Leaflet to create a custom web-based interface. The final design focuses on a one-screen overview for rapid comprehension, while State Profile and Compare modes support deeper analysis without cluttering the main dashboard.

The most important design lesson from the project is that adding more views and more resizing options does not necessarily improve usability. A dashboard should be structured around user tasks and visual hierarchy. By reducing duplicate sections, removing unsafe free resizing, keeping focus mode only for the map, and moving secondary analysis into separate modes, the final design becomes more stable and closer to dashboard design principles discussed in the course.

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A Appendix A: Evaluation questionnaire template

1. I could understand the purpose of the dashboard quickly. (1–5)
2. I could find the highest-pollution state for a selected year. (1–5)
3. I understood the difference between AQI and mean concentration. (1–5)
4. The map colours were understandable. (1–5)
5. The Top 10 cities chart was readable. (1–5)
6. The State Profile mode helped me understand the selected state. (1–5)
7. The Compare mode helped me compare two states. (1–5)
8. What was confusing?
9. What was the most useful part?
10. What should be improved?

B Appendix B: Additional running notes

The dashboard is submitted as a local Vite web application. The evaluator should first install dependencies with `npm install` and then start the dashboard with `npm run dev`. If the browser does not refresh after replacing files, a hard refresh with `Ctrl+Shift+R` clears cached assets. The production build can be generated with `npm run build`. The default entry point is the local URL printed by Vite, typically `http://localhost:5173/`.